

Mini Threat Intelligence Data Warehouse

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The project

An end-to-end data warehousing workflow

Integrate threat intelligence feeds for local SQL analysis

2 sources

URLhaus + ThreatFox

3 types

URL · domain · IP

1 data mart

Observation data

ACQUIRE → RECONCILE → MODEL → ANALYZE

Preserve source records · Apply common rules

Load a dimensional model · Run OLAP queries

Motivation and sources

Different formats and coverage; one analytical model

URLhaus

CSV feed

Malicious URLs

Status, tags and context

ThreatFox

JSON API + authentication

Multiple indicator types

Recent records: 1–7 days

Project scope: URLs, domains and IP addresses

Hash indicators are excluded during reconciliation

Reconcile once; analyze locally on the collected data

Architecture

Python ETL + PostgreSQL relational OLAP

01

Raw staging

Source-specific tables · Original records + ingest metadata
Append-only storage for traceability

02

Canonical layer

Common fields · Supported types · Conservative normalization
Deduplication: source + type + value + event day

03

Dimensional warehouse

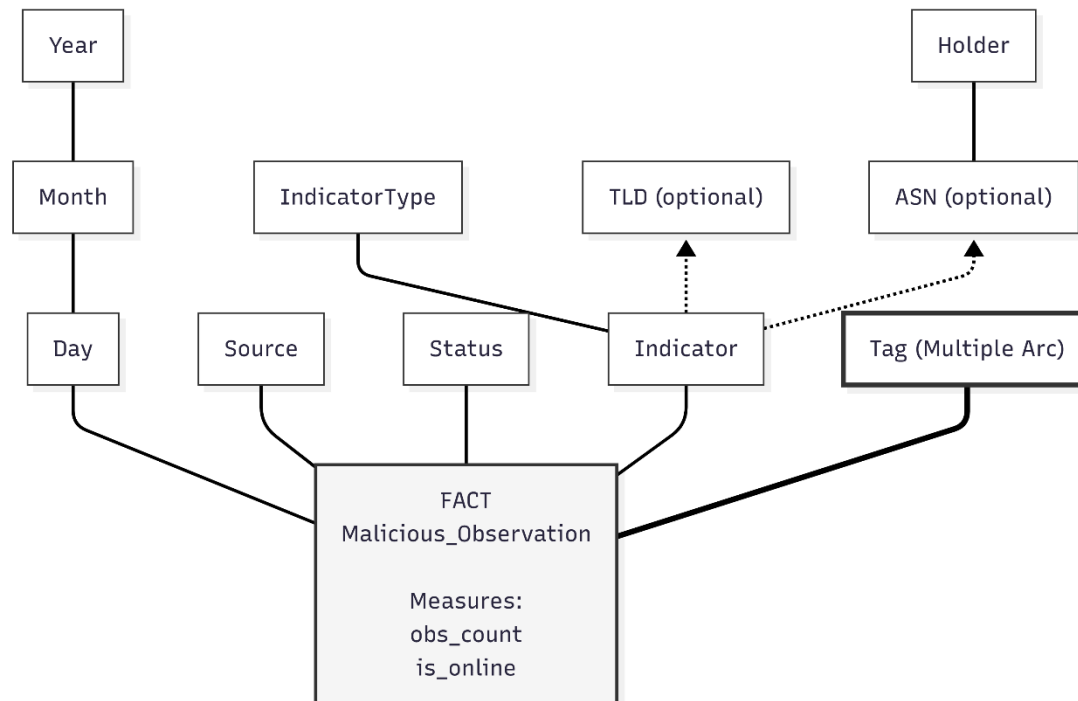
Dimension tables + observation fact + tag bridge
Upserts and unique keys prevent duplicate fact rows

URLhaus / ThreatFox → raw → canonical → DW → SQL

Conceptual model

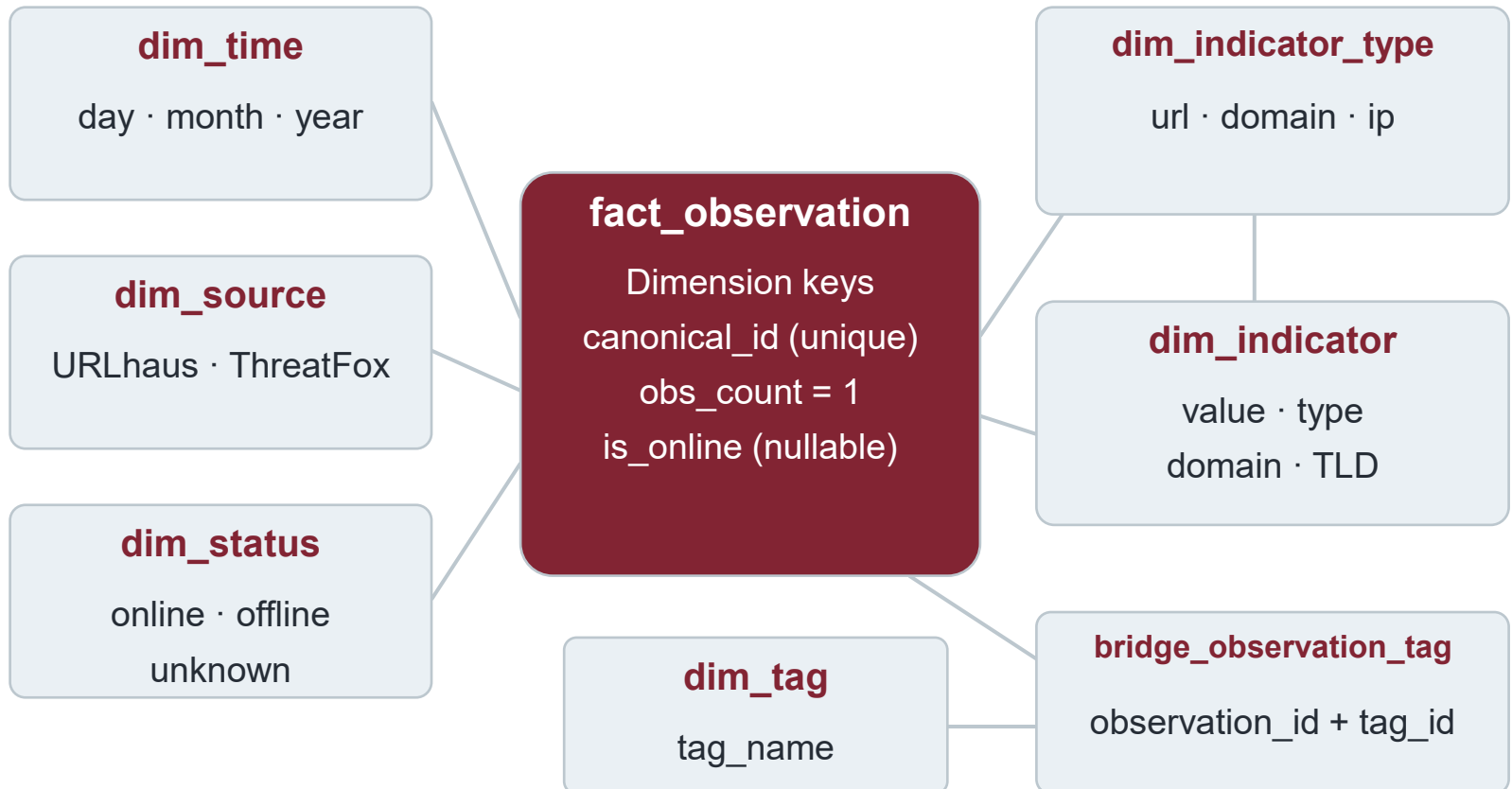
DFM: grain, measures and dimensional relationships

Grain: indicator (type + value) × source × event day



Star schema

Implemented PostgreSQL tables and relationships



Tags are many-to-many: avoid counting a fact again for each joined tag

OLAP queries

From analytical questions to SQL aggregations

MONTHLY TRENDS

How do volumes change over time?

Roll up day → month; SUM(obs_count) by source

SOURCE COMPOSITION

Which types does each source report?

Source + type; counts and within-source percentages

CROSS-SOURCE OVERLAP

Which indicators occur in both sources?

Exact (type, value) matches; top common indicators

TOP TLDs

Which TLDs are most frequent?

Rank the top 10 per source; exclude missing TLDs

Measure: collected feed observations